

If-else and loop Statements



The if Statement

- The **if** statement lets a program carry out different actions depending on a condition

```
if (balance >= price)
{
    purchase();
    balance = balance - price;
}
```

If / else Statement

```
if (balance >= price)
    { purchase();
      balance = balance - price;
    }
else
    { overdraft();
      purchase();
    }
```

If / else Statement

- *if(condition)*
statement
- *if (condition)*
statement1
else
statement2

Comparing Values: Relational Operators

Java	Math Notation	Description
>	>	Greater than
>=	≥	Greater than or equal
<	<	Less than
<=	≤	Less than or equal
==	=	Equal
!=	≠	Not equal

Multiple Alternatives: Sequences of Comparisons

- **if** (*condition1*)
 statement1;
 else if (*condition2*)
 statement2;
 ...
 else
 statement4;

Multiple Alternatives: Nested Branches

- Branch inside another branch

```
if (condition1)
{
    if (condition1a)
        statement1a;
    else
        statement1b;
}
else
    statement2;
```

Using Boolean Expressions: Predicate Method

- A predicate method returns a boolean value

```
public boolean isOverdrawn()
{
    return balance < 0;           // return (balance < 0);
}
```
- Use in conditions

```
if (Check.isOverdrawn()) . . .
```


Using Boolean Expressions: Predicate Method

- Useful predicate methods in **Character** class:
isDigit(Char c)
isLetter(Char c)
isUpperCase(Char c)
isLowerCase(Char c), isSpace(Char c)
- if (Character.isUpperCase(c)) . . .
- Useful predicate methods in **Scanner** class: hasNextInt() and hasNextDouble()
if (in.hasNextInt()) n = in.nextInt();

Using Boolean Expressions: The Boolean Operators

- `&&` and
- `||` or
- `!` not
- `if (0 < amount && amount < 1000) . . .`
- `if (input.equals("S") || input.equals("M")) . . .`

Self Check

- When does the statement `system.out.println (x > 0 || x < 0);` print false?

Answers

- When x is zero

while Loops

- Executes a block of code repeatedly
- A condition controls how often loop is executed
while (condition)
statement;
- Most commonly, the statement is a block statement (set of statements delimited by { })
while (condition) { statement; }

while Loop Flowchart

*while (condition)
statement;*

How to print int less then 10?

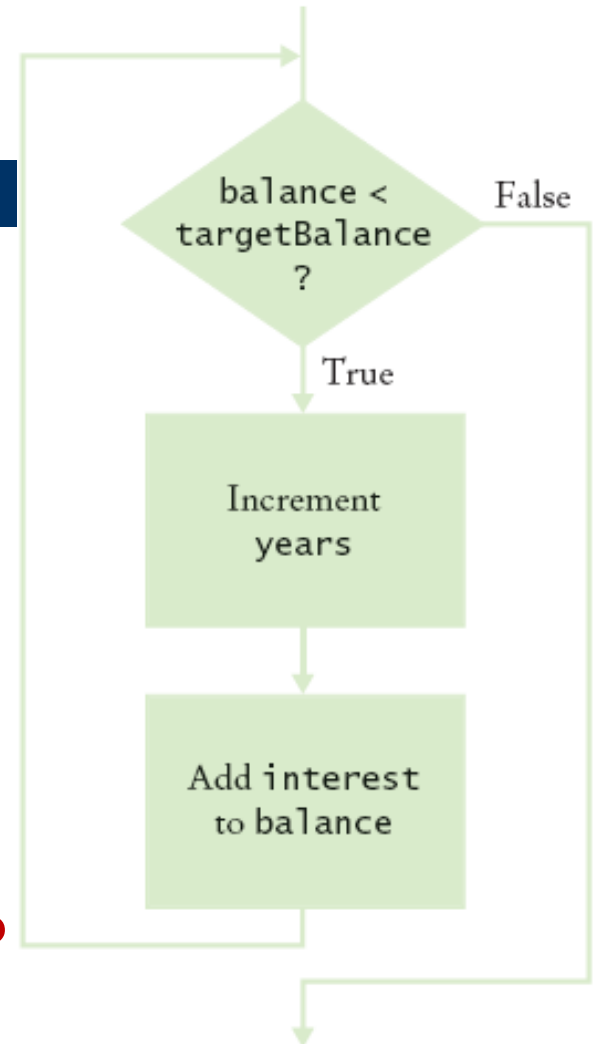


Figure 1 Flowchart of a while Loop

Self Check

- How many times is the statement in the loop *while (false) statement;* executed?

Answers

- Never

What if false condition is replaced by 0, 2, 10, 100, -5 , -1?

Common Error:

- ```
years = 0;
while (years < 20)
{
 interest = balance * rate / 100;
 balance = balance + interest;
}
```

 // should have years++
- ```
years = 20;
while (years > 0)
{
    years++;
    interest = balance * rate / 100;
    balance = balance + interest;
}
```

 // should have years--
- Loops run forever—must kill program

Do-while Loop

- Executes loop body at least once:

```
do{  
    statement;  
}  
while (condition);
```


Do-while Loop

Example: Validate input

- **double value;**
do
 {
 System.out.print("Please enter a positive number: ");
 value = in.nextDouble();
 }
while (value <= 0);

Alternative:

- **boolean done = false;**
while (!done)
 {
 System.out.print("Please enter a positive number: ");
 value = in.nextDouble();
 if (value > 0) done = true;
 }